

From Forecast Accuracy to Operational Value: Public Signal Augmentation for NYC 311 Service Demand

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Abstract

Day-ahead forecasts of municipal service-request volume are a natural input to staffing and dispatch planning, yet a reduction in average forecast error does not automatically translate into better operational decisions. We ask, on real data, two questions: do public external signals improve next-day forecasts of NYC 311 service-request volume, and does any improvement carry through to a stylized staffing-allocation simulation? Using real NYC 311 Service Requests for 2022-2024 (9,851,452 raw records aggregated to 43,240 date $\times$ borough $\times$ complaint-group observations) and real NOAA Central Park daily weather, we compare four feature sets—internal history, calendar, weather, and calendar + weather—across four model families under a strictly chronological protocol with five-fold rolling-origin validation. Calendar features deliver the largest accuracy gain across models; adding real weather further improves the selected random-forest model. The best model, a random forest on the calendar + weather feature set, attains a held-out test mean absolute error (MAE) of 55.325 versus 71.848 for a naive seasonal baseline, a 23.0% reduction. A stylized proportional staffing simulation shows the augmented policy reduces weighted unmet demand at all three tested crew budgets and closes 26.381% of the gap to an oracle at the baseline budget, but the magnitude of the decision benefit is budget-dependent. The central finding is that the calendar-augmentation gains are observed consistently across the evaluated splits, folds, boroughs, and complaint groups, while the decision gains are smaller and conditional on capacity. This study provides a reproducible evaluation of public-signal augmentation for municipal service-demand forecasting; no synthetic data or synthetic weather is used.

1. Introduction

Municipal service operations plan finite crews against demand that varies by day of week, season, and location. Short-horizon (next-day) volume forecasts are a standard planning input. Two questions follow. First, a predictive question: does augmenting an internal-history forecasting model with public external signals reduce next-day forecast error? Second, a decision question: does any such forecast improvement actually improve the quality of a capacity allocation built on top of the forecast?

These questions are related but distinct. Mean absolute error weights all errors equally across cells and days. A capacity allocation does not: under a fixed, scarce budget, an error on a high-volume or high-priority cell is more consequential than an error on a quiet one, and a proportional allocation maps forecasts to capacity non-linearly. A study reporting only forecast metrics can therefore misstate operational relevance.

We make three contributions. (i) A reproducible, real-data baseline on NYC 311 augmented with real NOAA weather, with a leakage-controlled chronological protocol. (ii) A controlled four-way feature-set comparison isolating the marginal value of calendar features and of real weather. (iii) A transparent forecast-to-decision simulation that measures whether forecast gains transfer to a stylized allocation, showing directionally positive but budget-dependent decision benefits. The study is predictive and correlational; Section 8 states the scope of the analysis and the natural directions for extending it.

2. Related Work

The methodological template is the demonstration that publicly available signals can improve operational forecasts. Cui et al. (2018) show, in a retail setting, that public social-media information improves daily sales forecasts across multiple machine-learning methods, with tree-based methods extracting more value from heterogeneous public signals than linear methods, and they frame the contribution as feature value rather than method novelty. Our work is in this lineage but moves from retail to municipal service operations, uses public weather rather than social media, and adds an explicit forecast-to-decision layer rather than stopping at forecast accuracy.

A second line argues that predictive accuracy and decision quality are distinct objectives. Elmachoub and Grigas (2022) develop a “predict, then optimize” framework in which models are trained with respect to the downstream optimization objective rather than prediction error alone, and Bertsimas and Kallus (2020) show how predictive models can be turned into prescriptions for decision problems. Our forecast-to-decision simulation is motivated by the same distinction, although we keep the allocation deliberately stylized and evaluate the decision objective rather than re-training the forecaster for it.

Within operations, public weather data has been shown to improve demand forecasts: Steinker, Hoberg, and Thonemann (2017) quantify the value of weather information for e-commerce order forecasting, which parallels our use of NOAA weather as a public signal. Planning capacity against time-varying demand is a long-standing service-operations problem; Green, Kolesar, and Whitt (2007) study staffing requirements when demand varies over time, motivating our crew-budget sensitivity analysis.

Prediction-driven prioritization in service and scheduling settings has been studied from an equity and efficiency angle (Samorani et al., 2022). We make no equity or causal claim; we cite this line only to situate the general point that the objective placed on top of a predictive model, not the model alone, shapes outcomes—which motivates our separate evaluation of the decision layer. Relatedly, our use of 311 data inherits a documented reporting bias: Kontokosta and Hong (2021) show that socio-spatial disparities in 311 complaint behaviour can make data-driven decisions unfair, which is why we frame the forecasting target as reported request volume rather than underlying need.

3. Data

Two real public sources are used; no synthetic data and no synthetic weather are used anywhere. Table 1 summarizes both.

NYC 311 Service Requests (NYC Open Data, Socrata dataset erm2-nwe9, “311 Service Requests from 2020 to Present”; City of New York, accessed 2026), restricted to 2022-2024. From 9,851,452 raw records the preprocessing procedure yields a 43,240-row date $\times$ borough $\times$ complaint-group panel spanning observed dates 2022-01-15 to 2024-12-30 (after lag and rolling warmup). Records are mapped to eight complaint groups (Housing, Noise, Other, Public Safety, Sanitation, Street Condition, Traffic, Water) by a deterministic substring rule and aggregated to daily counts over the five NYC boroughs. The forecasting target is the observed next-day request count per cell, defined only when consecutive calendar days are present.

NOAA NCEI Daily Summaries (GHCN-Daily) (NOAA NCEI, accessed 2026), station USW00094728 (NY City Central Park), 2022-2024, 1,096 daily rows. Variables used: precipitation, daily maximum and minimum temperature, average temperature, snowfall, snow depth, and wind speed. The source average-temperature field was empty for this station, so average temperature is derived as the mean of observed daily maximum and minimum; five missing wind-speed days are filled by time interpolation of neighbouring real observations. A single Central Park station is used as a city-level proxy, a documented spatial-resolution limitation.

Table 1. Dataset summary.

Property	Value
Service data	NYC 311 Service Requests (NYC Open Data, erm2-nwe9)
Study window	2022-01-01 to 2024-12-31
Raw records	9,851,452
Processed rows	43,240 (date $\times$ borough $\times$ complaint-group)
Observed range (processed)	2022-01-15 to 2024-12-30
Boroughs / complaint groups	5 / 8
Target	Observed next-day request count per cell
Weather source	NOAA NCEI Daily Summaries (GHCN-Daily)
Weather station	USW00094728 (NY City Central Park); single-station proxy
Weather rows / variables	1,096 / 7
Synthetic data or weather	none

4. Methods

We compare four feature sets: internal history (lag and rolling statistics of the per-cell series), calendar (internal plus deterministic calendar features), weather (internal plus the seven real weather variables), and calendar + weather (internal plus calendar plus weather). The corresponding configuration identifiers used in the results tables are `internal_historical`, `calendar_augmented`, `weather_augmented`, and `calendar_weather_augmented`. Full feature definitions are in Appendix A.

Four model families are evaluated: a naive seasonal baseline (trailing 7-day mean), Ridge regression, a random forest, and gradient boosting. Categorical features (borough, complaint group) are one-hot encoded; numeric features pass through. Lag and rolling features use only past observations, with rolling statistics shifted by one day to avoid same-day or future

leakage. Model hyperparameters are listed in Appendix I.

Validation uses a strictly chronological split by date: the earliest 70% of dates for training (30,240 rows), the next 15% for validation (6,480 rows), and the latest 15% for test (6,520 rows). The primary metric is MAE; RMSE, MAPE (computed with the denominator floored at one request, so that near-zero actual counts do not inflate the percentage error), and R^2 are also reported. The best configuration by validation MAE is refit on train + validation before the final test evaluation. Separately, five expanding-window rolling-origin folds assess stability.

The decision layer is a stylized staffing-allocation simulation. Each day a fixed crew budget is distributed across cells in proportion to the forecast using a largest-remainder rule. Each crew handles a fixed number of requests; unmet demand in a cell is the positive part of actual demand minus allocated capacity; weighted unmet demand assigns higher weight to Public Safety, Water, and Traffic. Policies driven by each feature set are compared against an oracle that allocates on the observed (realized) next-day request volume, which provides an upper bound only. Full assumptions are in Appendix D.

5. Forecasting Results

Calendar augmentation improves test MAE for every model. Adding real weather on top of calendar features further lowers error for the random forest, the selected model family; for Ridge and gradient boosting it does not improve on calendar alone (Table A3). For the random forest, MAE falls from 65.282 (internal history) to 58.082 (calendar); weather alone improves the internal-history model (64.104 versus 65.282), and calendar + weather is best, at 56.224 for the single-split model fit on the training set only. Ridge does not benefit from weather (internal 69.219, weather 69.283, calendar + weather 68.915). Figure 1 shows the best test MAE per feature set; the full per-model results are in Table 2 (selected rows are reported below; the complete table is in Appendix G).

Table 2 (selected rows). Held-out test MAE by model and feature set; lower is better. Full 13-row table in Appendix G.

Model	Feature set	Test MAE
naive_seasonal	naive_seasonal	71.848
random_forest	internal_historical	65.282
random_forest	calendar_augmented	58.082
random_forest	weather_augmented	64.104
random_forest	calendar_weather_augmented	56.224
gradient_boosting	calendar_weather_augmented	61.878
ridge	calendar_weather_augmented	68.915

The selected configuration by validation MAE is the random forest on calendar + weather. Refit on train + validation and evaluated once on the held-out test set, it attains test MAE 55.325, RMSE 177.476, MAPE 27.469%, and R^2 0.648—a 23.0% MAE reduction over the naive seasonal baseline (71.848). The selected-model test MAE of 55.325 (refit on train + validation) is therefore slightly lower than the 56.224 obtained from the train-only fit on the single split. Figure 2 shows observed versus predicted city-wide next-day totals over the test period, which track the overall level and weekly pattern of demand.

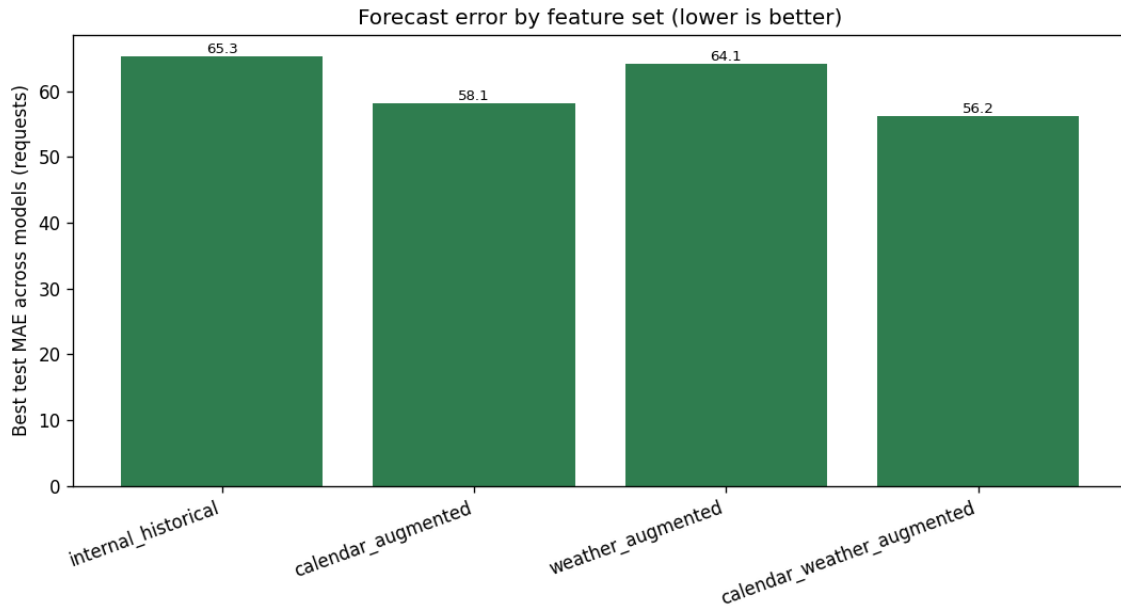


Figure 1. Best held-out test MAE achieved per feature set across models.

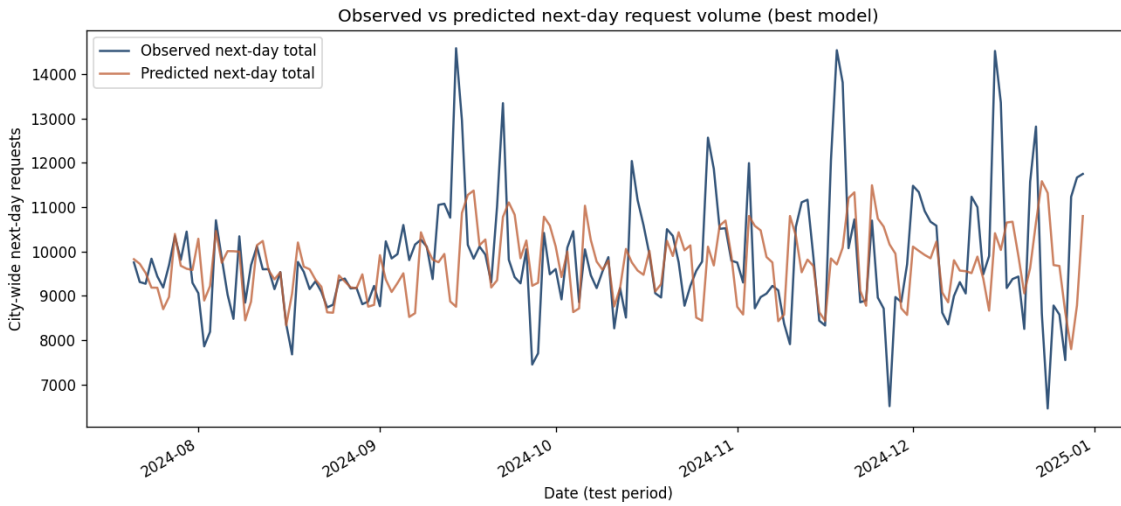


Figure 2. Observed versus predicted city-wide next-day request totals over the test period for the selected model.

6. Forecast-to-Decision Simulation

At the baseline budget (135 crews, 50 requests per crew, 163 test days), total weighted unmet demand is 643,324 under the internal-history policy and 631,191 under calendar + weather, versus an oracle of 597,332: the augmented policy reduces weighted unmet demand by 1.886% and closes 26.381% of the baseline-to-oracle gap.

Across crew budgets (Table 3, Figure 3) the calendar + weather policy is better than the baseline in all three settings, but the magnitude grows with capacity: the weighted-unmet reduction is 0.38% (scarce, 100 crews), 4.32% (moderate, 160), and 14.53% (generous, 220). The headline contrast—an approximately 11-15% forecast-accuracy gain versus a roughly

1.9% decision-quality gain at the baseline budget—is the central observation: average accuracy and operational value are not the same, and a scarce budget with demand concentrated in a few large cells limits how much any forecast can help.

Table 3. Decision sensitivity by crew budget. Stylized simulation; total weighted unmet demand (lower is better); calendar + weather versus internal history.

Setting	Crews	Internal	Calendar+Weather	Oracle	Reduction	Oracle gap closed
scarce	100	941,817	938,211	928,491	0.38%	27.06%
moderate	160	454,073	434,469	367,553	4.32%	22.66%
generous	220	164,197	140,346	50,865	14.53%	21.05%

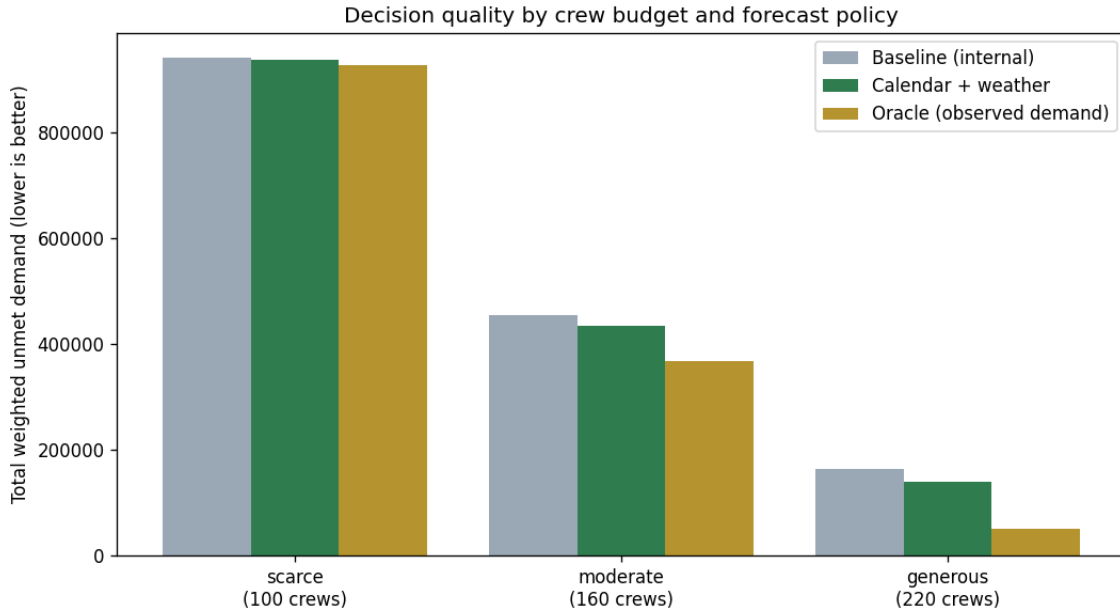


Figure 3. Total weighted unmet demand by policy under scarce, moderate, and generous crew budgets.

7. Robustness Checks

Five-fold expanding-window rolling-origin validation (Figure 4; full table in Appendix C) shows calendar augmentation lowering MAE in every fold for every model: mean improvement 15.13% for the random forest, 9.94% for gradient boosting, and 0.94% for Ridge, each in all 5 of 5 folds. The calendar + weather random forest has the lowest mean fold MAE (47.497). Segmented analysis (random forest, internal history versus calendar; Appendix G) shows the improvement holds for all five boroughs (about 8.8% to 18.8%) and all eight complaint groups (about 8.9% to 25.8%); absolute error concentrates in the highest-volume segments (Noise, Housing; the Bronx). The calendar improvement is therefore observed consistently across folds and segments, and is not driven by any single split.

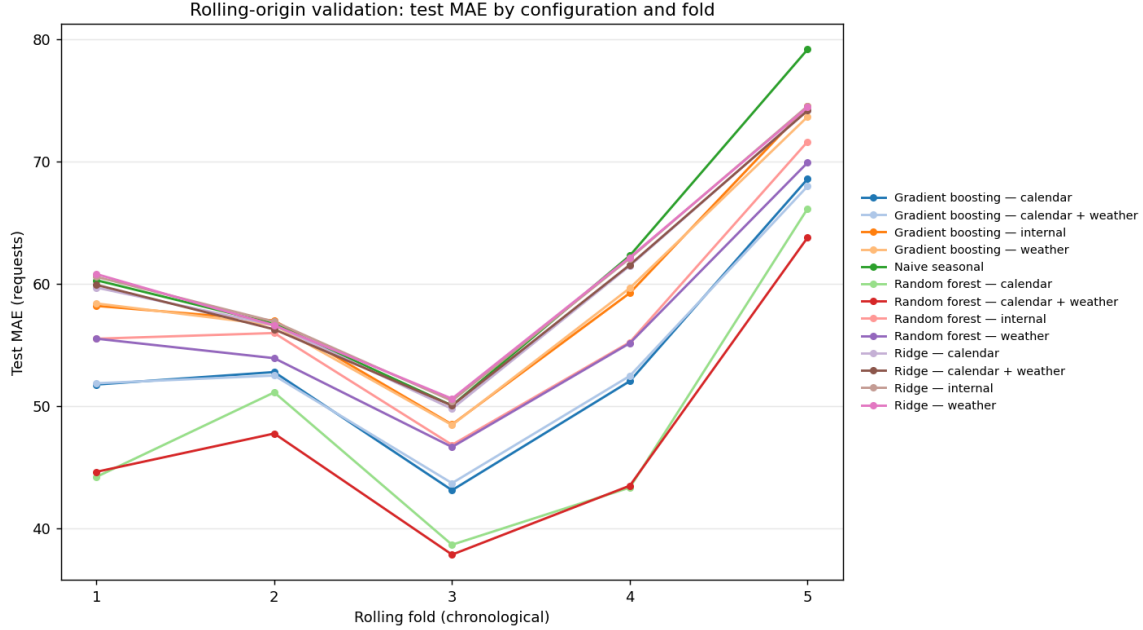


Figure 4. Per-fold test MAE by configuration across five expanding-window chronological folds.

8. Limitations

The study is predictive and correlational and makes no causal claim. NYC 311 reflects reporting behaviour, not true incidence (reporting bias), so forecasts predict reported volume. Weather is a single Central Park station used as a city-level proxy. Borough-level aggregation hides within-borough variation; the complaint-group mapping is a deterministic approximation with a residual Other group. No transit or event data is included. The staffing simulation is a stylized proportional heuristic—it omits crew travel, shifts, backlog, intra-day timing, and substitution, and it uses no observed dispatch decisions. Results apply to NYC and 2022-2024 only.

Natural extensions of this work: a non-stylized, agency-grounded decision model with an empirical service-level realism check; formal forecast-difference testing (for example a Diebold-Mariano test) with uncertainty intervals; and broader external signals beyond a single weather station, or cross-city replication.

9. Conclusion

On real NYC 311 data augmented with real NOAA weather, public-signal augmentation improves next-day service-demand forecasts—calendar features most, across rolling folds, boroughs, and complaint groups, with real weather adding a further gain for the selected random-forest model. The improvement transfers in direction to a stylized staffing allocation but with attenuated, budget-dependent magnitude. The calendar-augmentation gains are observed consistently across the evaluated splits, folds, and segments; the decision gains are smaller and conditional on capacity. This provides a reproducible evaluation framework for forecast-to-decision analysis in municipal service operations.

Reproducibility Statement

All reported values derive from the analysis described in Sections 3-7, implemented in Python 3.11 with fixed random seeds. The full procedure—weather ingestion, dataset construction, model training and evaluation, the decision simulation, rolling-origin validation, per-segment robustness analysis, and the crew-budget sensitivity analysis—is deterministic and reproducible from the two public data sources described in the references.

Data Availability Statement

NYC 311 Service Requests are publicly available through NYC Open Data (Socrata dataset `erm2-nwe9`), and NOAA NCEI Daily Summaries (GHCN-Daily) for station USW00094728 are publicly available through NOAA NCEI; full identifiers and retrieval dates are given in the references. The study window is 2022-2024. The aggregated daily counts and the derived weather series used in the analysis are obtained from these public sources following the procedure described above, and the full modelling dataset is reproducible from them. No synthetic data are used.

Code and Materials Availability

The analysis code, the derived summary outputs, and the figures reproduced here are available in the project repository at <https://github.com/Nassar-Coding/Nassar-Coding.github.io> (the public-signal-service-forecasting directory).

Ethics Statement

The study uses only public data at an aggregate date $\times$ borough $\times$ complaint-group level; no individual-level or personal data are used or produced, the analysis makes no individual-level decisions, and it is not deployed. Because NYC 311 reflects reporting behaviour rather than true incidence, and reporting propensity varies across communities and over time, forecasts of 311 volume describe reported demand rather than underlying need and should be interpreted accordingly.

Appendix

A. Feature and feature-set definitions

Lag features (request_lag_1, request_lag_7) and rolling statistics (rolling_mean_7/14, rolling_std_7/14) per (borough, complaint group) cell, using only past observations with a one-day shift. Calendar features: is_weekend, is_holiday, day_of_week, month, quarter, year, day_of_year, week_of_year, is_month_start, is_month_end. Weather features: precipitation_mm, temp_max_c, temp_min_c, temp_avg_c, snowfall_mm, snow_depth_mm, wind_speed_ms. The four feature sets—internal history (internal_historical), calendar (calendar_augmented), weather (weather_augmented), and calendar + weather (calendar_weather_augmented)—are listed in Table A1.

Table A1. Feature sets.

Feature set	Features
internal_historical	borough, complaint_group, request_lag_1, request_lag_7, rolling_mean_7, rolling_mean_14, rolling_std_7, rolling_std_14
calendar_augmented	internal_historical + is_weekend, is_holiday, day_of_week, month, quarter, year, day_of_year, week_of_year, is_month_start, is_month_end
weather_augmented	internal_historical + precipitation_mm, temp_max_c, temp_min_c, temp_avg_c, snowfall_mm, snow_depth_mm, wind_speed_ms
calendar_weather_augmented	internal_historical + calendar features + weather features

B. Complaint-group mapping

Raw complaint types are upper-cased and matched against an ordered list of substring rules; the first matching group wins, and any record matching none is assigned the residual Other group. The order places Housing before Water so that, for example, “HEAT/HOT WATER” is classified as Housing rather than Water. The matched substrings per group are as follows.

Noise: NOISE, LOUD. **Housing:** HEAT/HOT WATER, HEAT, HOT WATER, PLUMBING, PAINT, PLASTER, APPLIANCE, DOOR, WINDOW, ELECTRIC, FLOORING, STAIRS, ELEVATOR, MOLD, GENERAL CONSTRUCTION, HOUSING, APARTMENT, UNSANITARY CONDITION, OUTSIDE BUILDING. **Sanitation:** SANITATION, DIRTY, MISSED COLLECTION, LITTER, GARBAGE, RECYCLING, WASTE, DUMPING, GRAFFITI, RODENT, OVERFLOWING. **Street Condition:** STREET CONDITION, STREET LIGHT, POTHOLE, SIDEWALK, CURB, ROAD, STREET SIGN, TRAFFIC SIGNAL, HIGHWAY. **Water:** WATER, SEWER, HYDRANT, LEAK, FLOOD, DRAINAGE, CATCH BASIN. **Traffic:** ILLEGAL PARKING, BLOCKED DRIVEWAY, TRAFFIC, PARKING, ABANDONED VEHICLE, DERELICT VEHICLE, DRIVEWAY. **Public Safety:** ILLEGAL FIREWORKS, DRUG, WEAPON, ASSAULT, SAFETY, EMERGENCY, ENCAMPMENT, HOMELESS, ANIMAL ABUSE, DISORDERLY, URINATING, PANHANDLING.

C. Chronological split and rolling-origin setup

Chronological 70/15/15 split by date (30,240 / 6,480 / 6,520 rows). Five expanding-window rolling-origin folds; each fold trains on all dates up to a cut point and tests on the next block, with no future information in training. Table A2 reports the mean fold MAE that underlies Figure 4.

Table A2. Rolling-origin validation: random-forest mean fold test MAE by feature set.

Configuration	Mean MAE	Std
naive_seasonal	61.701	9.673
random_forest internal_historical	57.027	8.046
random_forest weather_augmented	56.222	7.561
random_forest calendar_augmented	48.690	9.590
random_forest calendar_weather_augmented	47.497	8.746

D. Decision-simulation assumptions

Each day, a fixed crew budget is allocated across the 40 (borough $\times$ complaint-group) cells in proportion to each cell's forecast, rounded by a largest-remainder rule with a minimum of zero crews per cell. Each crew serves 50 requests per day, so a cell's allocated capacity is 50 times its crew count. Unmet demand in a cell is $\max(0, \text{actual next-day requests} - \text{allocated capacity})$. Weighted unmet demand multiplies each cell's unmet demand by a complaint-group weight—Public Safety 2.0, Water 1.5, Traffic 1.5, and 1.0 for Noise, Sanitation, Street Condition, Housing, and Other—and is summed over cells and test days. The baseline-budget configuration uses 135 crews over 163 test days. The oracle allocates using the observed (realized) next-day request volume and provides an upper bound on achievable performance within this stylized simulation; it is not a forecast. The simulation uses no observed dispatch decisions.

E. Crew-budget sensitivity

Three budgets—scarce (100 crews, 5,000 daily capacity), moderate (160 / 8,000), and generous (220 / 11,000); per-policy results are reported in Table 3.

F. Weather-data treatment

Single NOAA station (USW00094728) joined to all boroughs by date; temp_avg_c derived from observed daily maximum and minimum (source average-temperature field empty); five missing wind-speed days time-interpolated from real neighbours; no synthetic weather. Figure A1 shows the series.

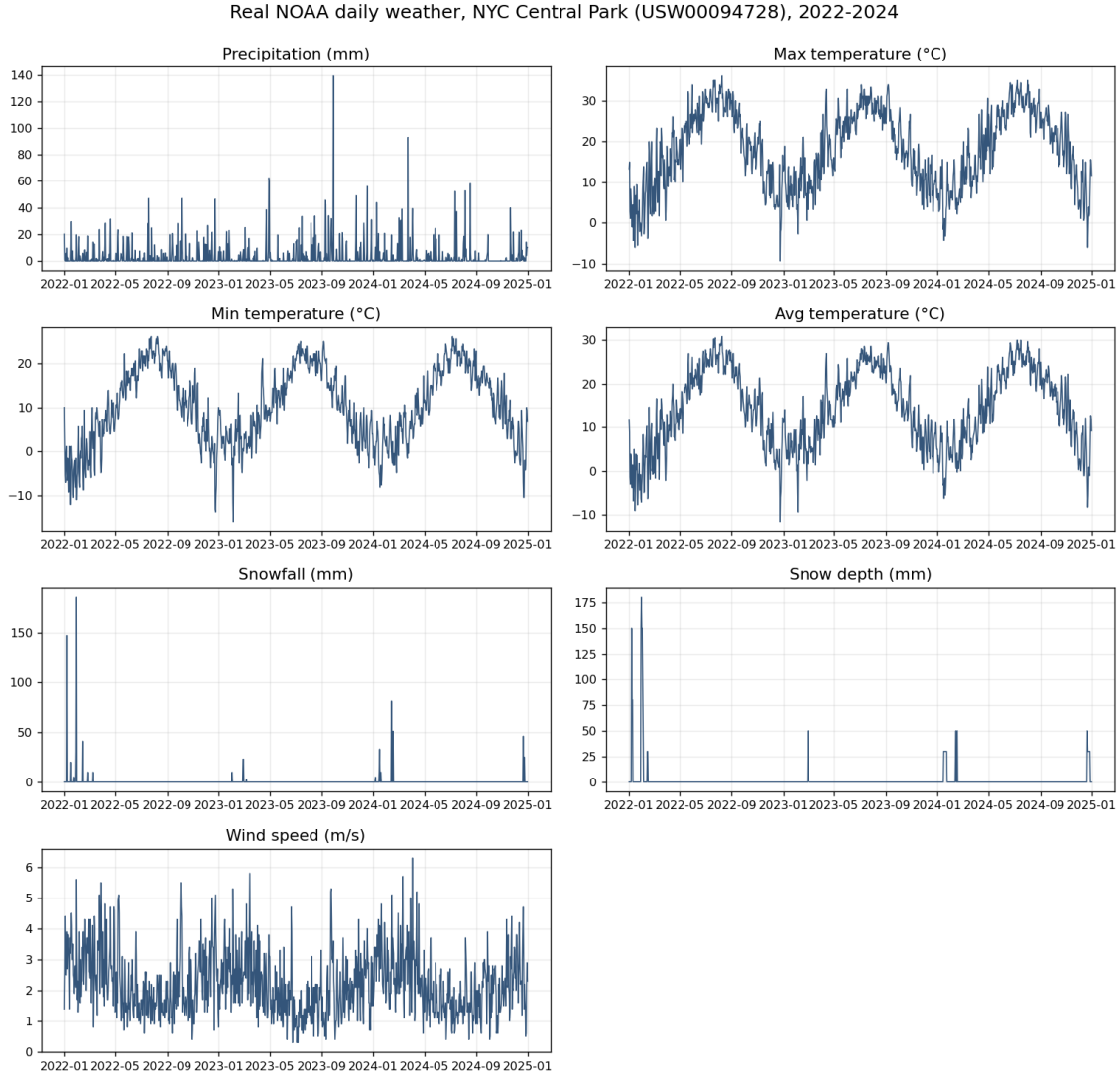


Figure A1. Real NOAA daily weather series for the Central Park station, 2022-2024.

G. Full results tables

Table A3 is the complete 13-row model comparison (Figure A2). Table A4 reports per-borough robustness (Figure A3) and Table A5 per-complaint-group robustness (Figure A4), each comparing internal history with calendar for the random forest. Figure A6 shows internal history versus calendar per model; Figure A5 shows a baseline-budget policy comparison with the oracle.

Table A3. Full model comparison (held-out test).

Model	Feature set	Test MAE	Test RMSE	Test MAPE (%)	Test R ²
naive_seasonal	naive_seasonal	71.848	197.743	34.689	0.563
ridge	internal_historical	69.219	187.753	36.073	0.606
ridge	calendar_augmented	68.758	187.757	35.729	0.606
ridge	weather_augmented	69.283	187.784	36.009	0.606
ridge	calendar_weather_augmented	68.915	187.762	36.140	0.606
random_forest	internal_historical	65.282	184.787	34.259	0.618
random_forest	calendar_augmented	58.082	181.135	29.026	0.633
random_forest	weather_augmented	64.104	181.379	33.614	0.632
random_forest	calendar_weather_augmented	56.224	178.150	28.173	0.645
gradient_boosting	internal_historical	67.493	187.734	38.378	0.606
gradient_boosting	calendar_augmented	61.710	183.886	34.152	0.622
gradient_boosting	weather_augmented	67.553	188.046	38.236	0.605
gradient_boosting	calendar_weather_augmented	61.878	181.778	33.822	0.631

Table A4. Borough robustness (random forest; internal history versus calendar).

Borough	Internal MAE	Calendar MAE	Improvement (%)
Bronx	120.194	109.567	8.84
Brooklyn	76.991	64.175	16.65
Manhattan	58.178	48.078	17.36
Queens	53.771	43.675	18.78
Staten Island	12.766	11.129	12.82

Table A5. Complaint-group robustness (random forest; internal history versus calendar).

Complaint group	Internal MAE	Calendar MAE	Improvement (%)
Noise	194.882	168.272	13.65
Housing	121.762	106.839	12.26
Other	54.794	44.474	18.83
Traffic	45.000	40.990	8.91
Sanitation	24.957	22.078	11.54
Public Safety	24.449	21.496	12.08
Water	25.111	20.585	18.02
Street Condition	24.084	17.865	25.82

Table A6. Limitations and mitigations.

Limitation	Mitigation / Scope statement
311 reporting bias	Documented; target framed as reported volume; no need-based or causal claim
Borough-level aggregation	Granularity stated; within-borough variation not captured
Complaint-group mapping coarseness	Deterministic, documented; residual Other group reported
Single-station weather proxy	Documented spatial-resolution limitation; city-level proxy
Derived average temperature	Mean of observed daily max/min; source field empty; documented
No transit or event data	Scope stated; future work
Stylized decision simulation	Labelled stylized; oracle bound included for context
No observed dispatch decisions	Counterfactual only; no real-dispatch or optimization claim
Forecast-versus-decision gap	Decision metrics reported; transfer attenuated and budget-dependent
No causal inference	No causal claim anywhere
NYC-only, 2022-2024	External-validity limitation stated explicitly

H. Reproducibility

The full analysis is reproducible from the public data sources following the protocol described in Sections 3-7; see the Reproducibility Statement.

I. Model configurations

All models are scikit-learn estimators trained with random seed 42. The naive seasonal baseline predicts the trailing 7-day mean. Ridge regression uses $\alpha = 1.0$. The random forest uses 200 trees, maximum depth 14, and a minimum of two samples per leaf. Gradient boosting uses 200 stages, maximum depth 3, and learning rate 0.05. All other hyperparameters take their scikit-learn defaults; no additional hyperparameter tuning was performed, and model selection chooses among feature sets by validation MAE.

Appendix Figures

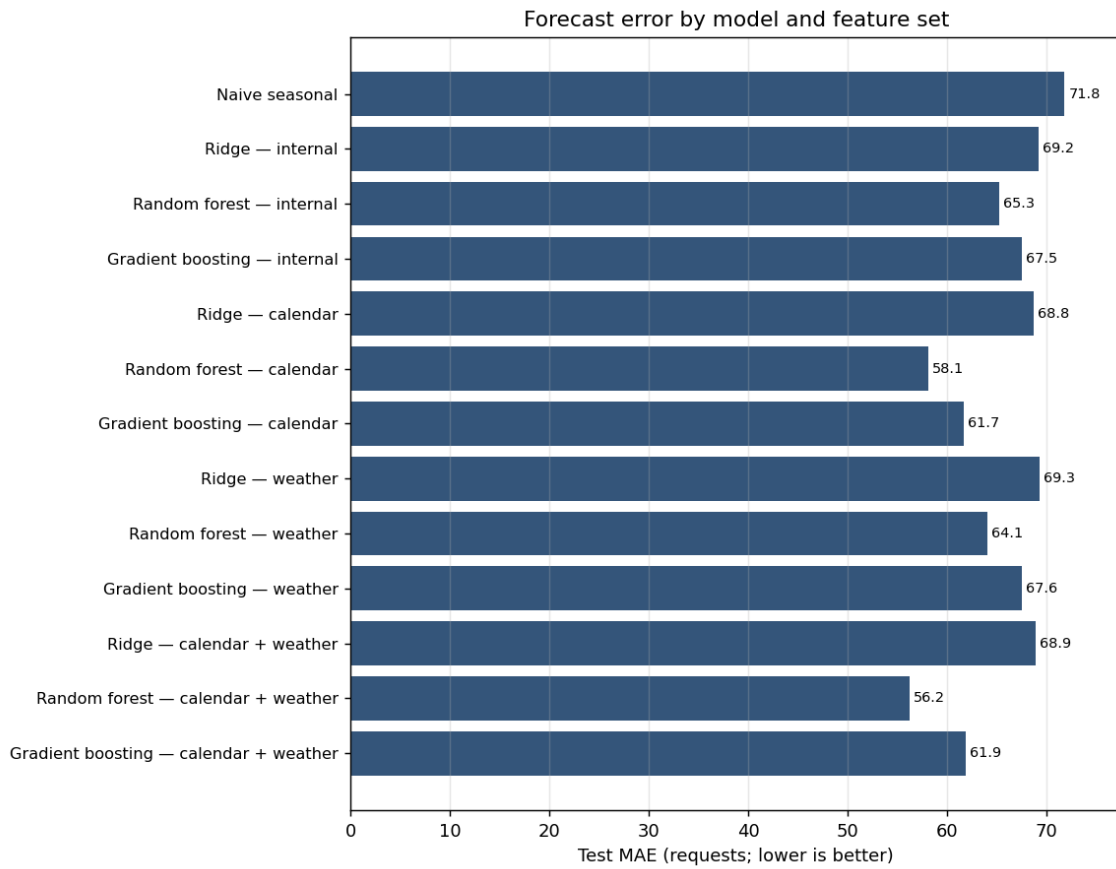


Figure A2. Held-out test MAE for the naive baseline and every model and feature-set combination.

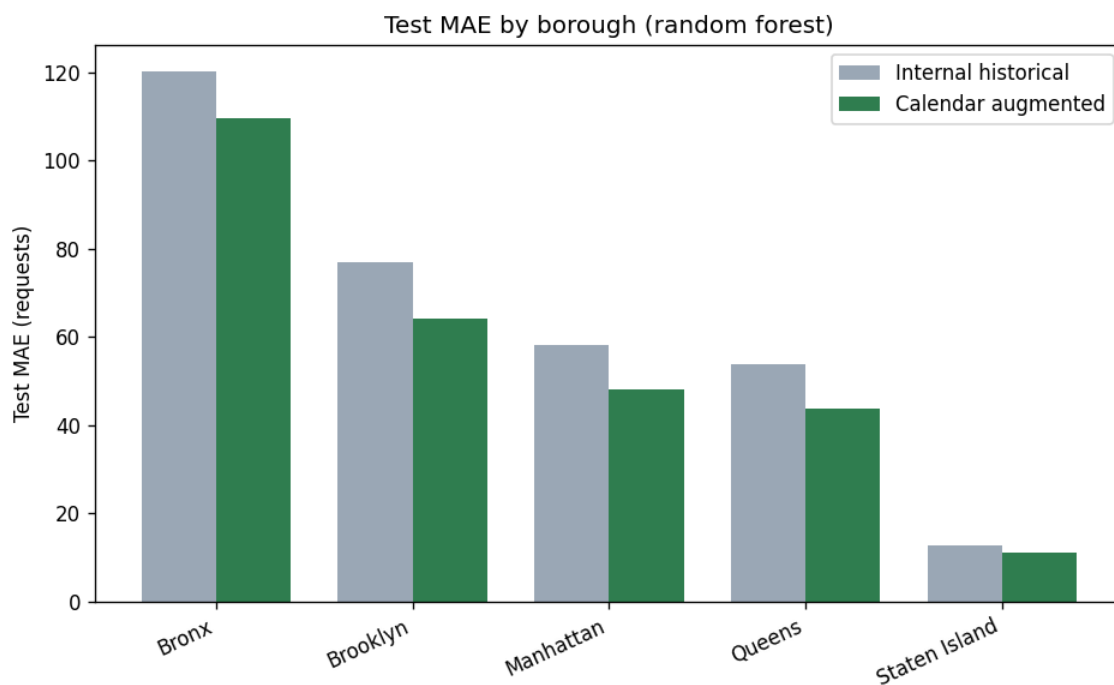


Figure A3. Internal history versus Calendar test MAE by borough (random forest).

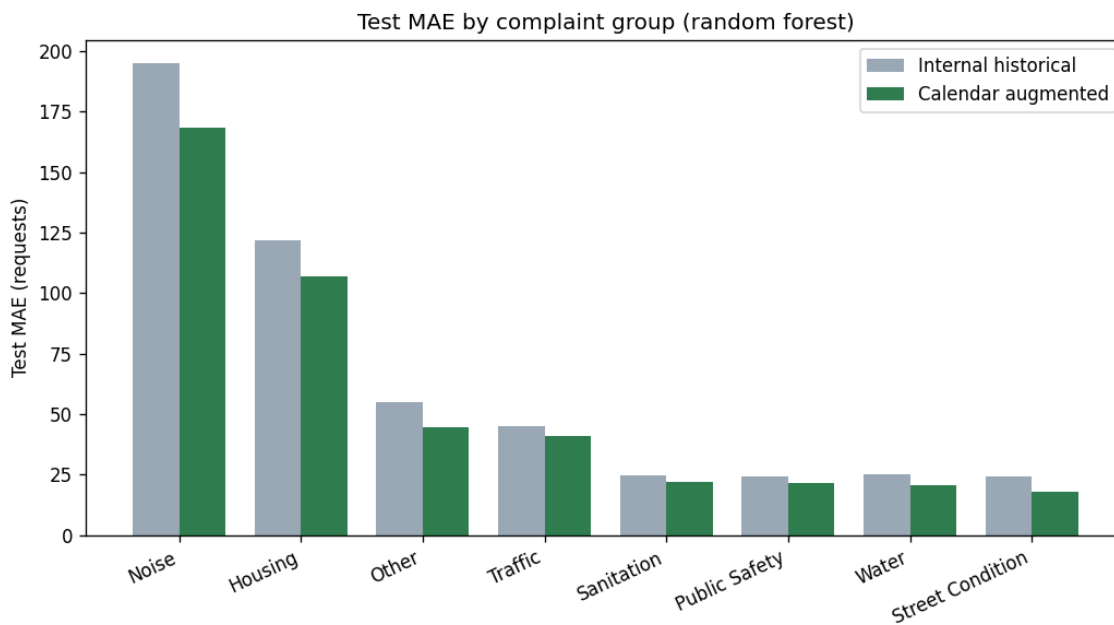


Figure A4. Internal history versus Calendar test MAE by complaint group (random forest).

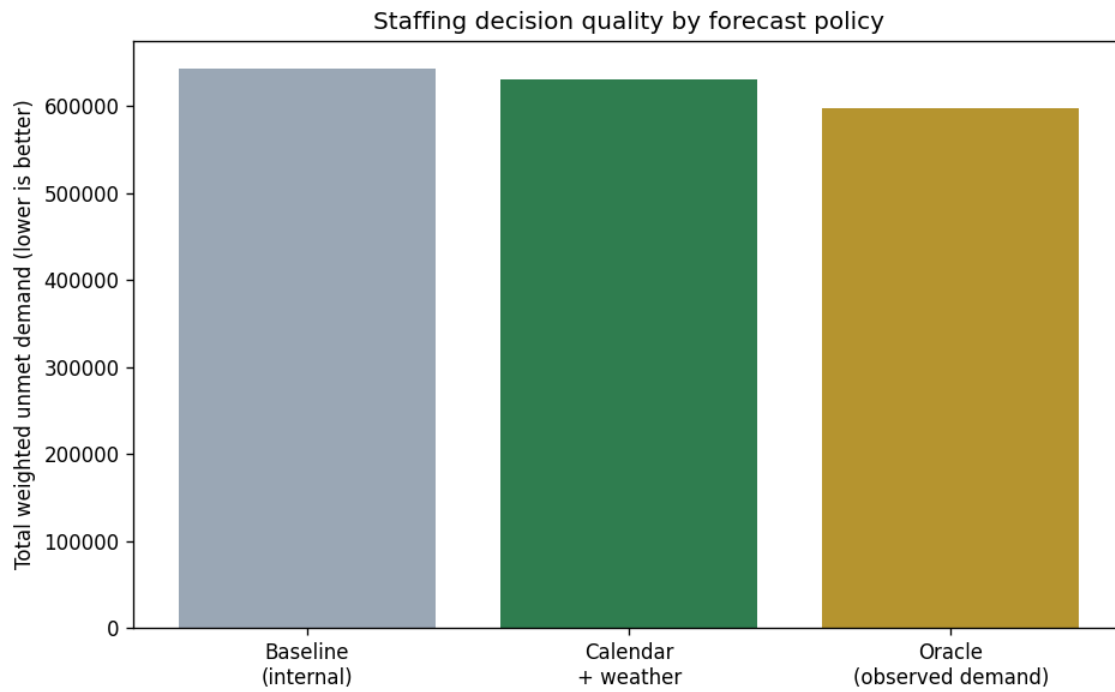


Figure A5. Total weighted unmet demand by policy at the baseline crew budget, with the oracle benchmark.

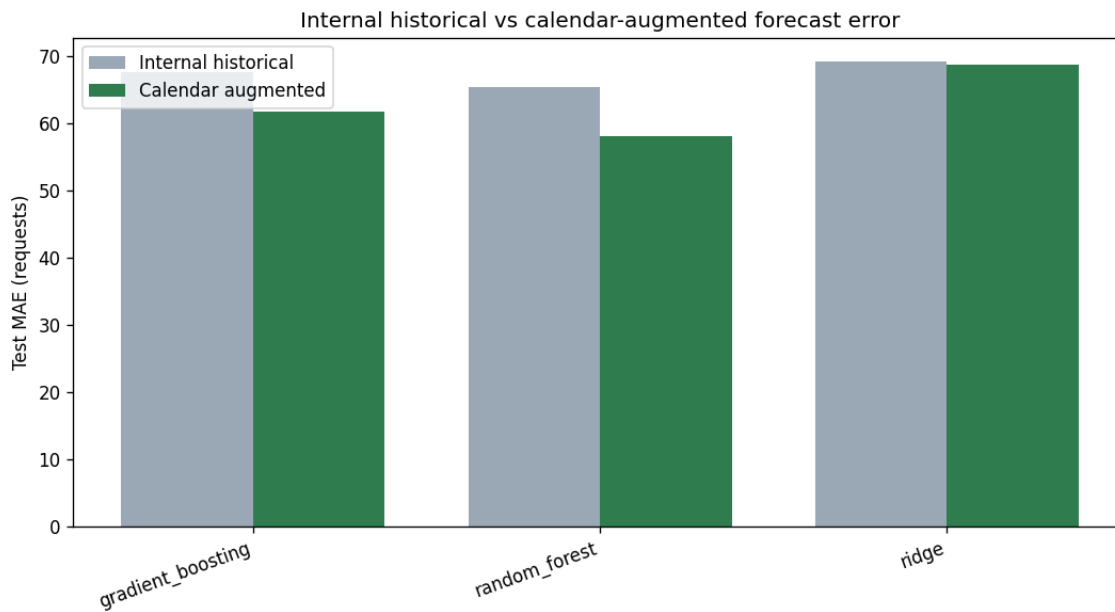


Figure A6. Internal history versus Calendar test MAE per non-naïve model.

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